VIBEMATCH

Semantic Vector-Based Matchmaking System

VIBEMATCH is an AI-powered matchmaking platform that combines semantic embeddings, multi-dimensional clustering, game-theoretic algorithms, and ensemble learning to deliver highly compatible matches. The system processes user data across 4 layers (music, identity, behavior, preferences) and dynamically assigns users to multiple overlapping sets for nuanced, scalable matching.

SYSTEM ARCHITECTURE

VIBEMATCH uses a **layered, multi-set hierarchical architecture** with quality-gated clustering:

```
Layer 1: Music Data (384-dim SentenceTransformer vectors)
       ↓
Layer 2: Identity Data (sex, gender, sexuality, location)
       ↓
Layer 3: Behavioral Data (NLP bio embeddings, interaction history)
       ↓
Layer 4: Preference Data (what they're looking for)
       ↓
Multi-Set Assignment (User in 4+ sets simultaneously)
       ↓
Hierarchical Clustering (Sets → Clusters with quality gates)
       ↓
Matching Pipeline: Delaunay → Score → Gale-Shapley → Ensemble
       ↓
Result: 20 candidates ranked by compatibility + stability
```

MULTI-SET CLUSTERING

Key Innovation: Users assigned to multiple overlapping sets (not single bucket). **Example - User A (22yo woman, NYC):**

Set Type	Set Name	Criteria	Likeability
Demographic	Set-D1	age: 20-22, woman, straight, NYC	0.72
Music	Set-M1	indie-pop, similarity > 0.80	0.85
Behavioral	Set-B1	active daters (>10 matches)	0.65
Preference	Set-P2	looking for outdoorsy types	0.78

Clustering: Sets grouped into higher-level clusters (e.g., "Young NYC Indie Lovers"). Quality gates automatically detect and remove bad sets (insufficient data, low match rate).

HARDCODED SCORING SYSTEM

Developer-Set Weights (Immutable):

```
MUSIC_COMPATIBILITY:    40%  (384-dim vector cosine similarity)
PERSONALITY_SCORE:      30%  (Layer 3 - Phase 3 activation)
BEHAVIOR_SCORE:         20%  (Layer 3 - Phase 3 activation)
ATTRACTION_SCORE:       10%  (Layer 4 - Phase 4 activation)

final_score = (music*0.40) + (personality*0.30) +
              (behavior*0.20) + (attraction*0.10)
```

7-STAGE MATCHING PIPELINE

Stage	Operation	Complexity
1	Set Assignment	O(1) hash lookup
2	Intersection & Clustering	O(n log n) Delaunay + DBSCAN
3	Coarse Filtering	O(n log n) spatial+vector clustering
4	Likeability Scoring	O(n) set-level scoring
5	Composite Scoring	O(n) hardcoded weighted sum
6	Gale-Shapley	$O(n^2)$ stable matching guarantee
7	Ensemble Ranking	O(n) GS+Behavior+ML ensemble

Gale-Shapley Guarantee: Produces stable matching where no user-pair would prefer each other over current partners. Eliminates one-sided crushes, increases mutual interest.

TIME-STAMPED ONBOARDING

Progressive Data Collection (Target: 20 minutes)

Time	Action	Layer	Richness	Status
T+0	User signs up	-	0%	INCOMPLETE
T+5min	Add music (20+ artists)	Layer 1	20%	MUSIC_ADDED
T+12min	Fill demographics	Layer 2	40%	IDENTITY_ADDED
T+20min	Write bio + prompts	Layer 3	65%	COMPLETE
T+20min	Final set assignment	-	65%	READY_TO_MATCH

ENSEMBLE MATCHING & FEEDBACK LOOPS

After 20+ swipes, system activates ensemble learning:

```
Component 1: Gale-Shapley Score          (40%)
Component 2: Behavioral Clustering       (30%)
Component 3: ML Match Prediction         (30%)

Result: Match success improves 35% → 68% over 3 months
```

Dynamic User Mobility: Users automatically transition between sets as behavior evolves (e.g., selective → active dater). System learns from interactions and refines rankings.

IMPLEMENTATION ROADMAP

Phase	Timeline	Key Features	Status
Phase 1	Now	Music embeddings, single scoring, basic sets	IN PROGRESS
Phase 2	Months 3-4	Multi-set, clustering, Gale-Shapley	PLANNED
Phase 3	Months 5-8	Personality/Behavior scores, Ensemble ML	PLANNED
Phase 4	Months 9-12	Face embeddings, Full ensemble, Optimization	PLANNED

TECHNICAL STACK

Component	Technology	Purpose
Backend	FastAPI (Python)	REST API, real-time matching
Embeddings	SentenceTransformer (all-MiniLM-L6-v2)	384-dim semantic vectors
Clustering	Delaunay triangulation + DBSCAN	Hierarchical spatial clustering
Matching	Gale-Shapley Algorithm	Stable matching guarantee
ML	XGBoost Ensemble	Match success prediction
Database	SQLite → PostgreSQL + pgvector	Vector similarity search
Frontend	React Native	iOS/Android app

KEY INNOVATIONS

1. **4-Layer Data Architecture:** Music, Identity, Behavior, Preferences for multi-dimensional matching
2. **Multi-Set Clustering:** Users in overlapping sets (flexible, not siloed)
3. **Quality Gates:** Automatic detection & removal of low-quality sets
4. **Game-Theoretic Matching:** Gale-Shapley ensures stable, mutual matches
5. **Hardcoded Scoring:** Predictable, developer-controlled weights
6. **Ensemble Learning:** Combines 3 approaches for 68%+ success rate
7. **Time-Stamped Onboarding:** Gradual data collection, sigmoid visibility ramp
8. **Dynamic Mobility:** Users evolve between sets based on behavior

EXPECTED PERFORMANCE (12-month projection)

Metric	Phase 1	Phase 2	Phase 3	Phase 4
Match Success	35%	48%	62%	68%
Avg Time to Match	2.5 min	1.8 min	1.5 min	1.2 min
Conversation Rate	25%	35%	52%	68%
User Retention	45%	58%	70%	78%
Scalability	100K users	500K users	2M users	10M+ users

VIBEMATCH - Proof of Concept
Generated: May 19, 2026
Repository: github.com/d3-f4u1t/VIBEMATCH
Contact: For questions on architecture, implementation, or deployment.